MIL-H-13528B
27 February 1981
SUPERSEDES
MIL-A-13528A
20 May 1974

MILITARY SPECIFICATION

HYDROCHLORIC ACID, INHIBITED, RUST-REMOVING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers inhibited hydrochloric acid for use in removing heavy rust deposits from steel surfaces.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-637 - Steel Bar, Carbon, Cold Finished (Standard Quality, Free Machining)
QQ-S-698 - Steel, Sheet and Strip, Low Carbon
PPP-C-1337 - Container, Composite, (Steel Drum With Polyethylene Insert)

STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data Sheets, Preparation and the Submission of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Materials and Mechanics Research Center, ATTN: DRXMR-LS, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized and Containerized Unit Loads, 40 Inch x 48 Inch Pallets, Skids, Runners or Pallet Type Base
MIL-STD-1188 - Commercial Packaging of Suppliers and Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF TRANSPORTATION, LAWS AND REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Code of Federal Regulations, CFR, and the Federal Register are for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 8 - Tension Testing of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Material. The material shall consist of a technical grade hydrochloric acid, of a normality concentration of 4.7 to 4.9, admixed with suitable inhibiting agents to reduce its metal corrosiveness characteristics.

3.2 Corrosiveness. The average weight loss of 1020 cold rolled steel panels, pickled at 77 F (25 C) and at 100 F (37.8 C) in the inhibited acid furnished under contract, shall not be greater than the average weight loss of similiar panels pickled in like manner in the reference standard inhibited hydrochloric acid (see 4.5.1). Furthermore, the test panels shall not be pitted by the acid.

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3.3 Hydrogen embrittlement-tensile effects. The average reduction in area of 1213 cold-drawn steel bars, pickled at 77 F (25 C) and at 100 F (37.8 C) in the inhibited acid furnished under contract, shall be equal to or greater than the reduction in area of similar test specimens pickled in like manner in the reference standard inhibited acid (see 4.5.1). However, if the reduction in area of the specimens pickled in the reference standard acid is greater than the reduction in area of the untreated control test specimens (see 4.5.3), the acid furnished under contract will be acceptable.

3.4 Rate of rust removal. The average time required by the inhibited acid furnished under contract to remove rust from 1020 cold rolled steel weather-rusted panels, shall be not longer than the average time required by the reference standard acid on similar specimens likewise tested.

3.5 Toxicity. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the procuring activity to the appropriate departmental medical service who will act as an advisor to the procuring agency. A Material Safety Data Sheet shall be furnished in accordance with FED-STD-313.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot of inhibited hydrochloric acid shall consist of not more than 1,000 gallons of material produced by one manufacturer in a batch process, and submitted at one time for acceptance.

4.3 Sampling.

4.3.1 Sampling for examination of filled containers. A random sample of filled containers shall be taken in accordance with MIL-STD-105 at inspection level 1 and acceptable quality level equals 2.5 percent defective to verify compliance with all stipulations of this specification regarding fill, closure, marking and other requirements not involving tests.

4.3.2 Sampling for tests. Two containers from each batch shall be selected at random and a one-gallon sample shall be taken from each container.

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4.4 Inspection

4.4.1 Examination of filled containers. Each sample filled container selected as in 4.3.1 shall be examined for defects of construction of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each filled container shall also be examined for required fill. Any container in the sample having one or more defects, or under required fill, shall be cause for rejection; and if the number of nonconforming containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105 the lot represented by the sample shall be rejected.

4.4.2 Lot acceptance tests. Each sample taken as in 4.3.2 shall be subjected to the tests specified in 4.5. Failure of either sample to comply with any of tests specified herein shall be cause for rejection of the lot or batch.

4.5 Test procedures

4.5.1 Reference standard inhibited hydrochloric acid. The reference standard inhibited acid used for comparison in the tests shall be prepared as follows:

Place 10.0 grams each of para-toluidine and hexamethylene tetramine in a 2,000-ml volumetric flask. Add 100 ml of 4.8N technical grade hydrochloric acid at room temperature. Let the mixture stand until a precipitate forms. Add approximately 800 ml of distilled water and 700 ml of concentrated technical grade hydrochloric acid to bring the final solution to a normality of 4.8, mixing the solution well, cooling it to room temperture, and bringing it up to volume with distilled water.

4.5.2 Corrosiveness

4.5.2.1 Panel preparation. Sixteen panels of cold-rolled and thoroughly annealed 18-gage 1020 steel, each 2-1/2 by 2-1/2 inches (6.4 by 6.4 cm) shall be prepared. The steel sheets or strips used to make the panels shall conform to QQ-S-698. The edges of the panels shall be milled square. All rough spots on the panel surfaces shall be removed with No. 1 emery cloth (approximately 100 strokes are normally required), followed by rubbing with crocus cloth (100 stroke on each face). Finally the faces of the panels shall be rubbed against analytical grade Celite (hydrated amorphous silica). The edges of the panels shall be treated in identical sequence, but using only half the number of strokes required for the panel faces. The panels shall then be swabbed with absorbent cotton under running water, washed with alcohol, and air-dried. The panels shall be handled with tweezers subsequent to the celite treatment, to avoid contact with the hands of the operator.

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4.5.2.2 Treatment procedure at 77 F (25 C). Eight of the test panels shall be individually weighed, to the nearest 0.1 milligram, and placed into separate petri dishes, 100 millimeters in diameter by 20 millimeters deep with each panel held by a glass hook so that only one of its edges touches the bottom of the dish. A sufficient amount of the standard reference acid at 77 F (25 C) shall be added to four of the dishes (approximately 45 ml to each dish), so as to completely cover the panel. A like amount of the inhibited acid under test, also at 77 F (25 C) shall be added to each of the other four dishes. Record the time of addition of the acids. After 90 minutes of pickling in the acids, maintained at 77 F ± 2 F (1.1 C) the panels shall be removed, washed thoroughly with water, dried with paper toweling, rubbed with analytical grade celite (50 times on each face), again washed with water, then with alcohol, air-dried, and reweighed. The average loss in weight of the two sets of panels shall be calculated, and compared. The panels shall then be observed under 6X magnification for evidence of pitting.

4.5.2.3 Treatment procedure at 100 F (37.8 C). The procedure for the pickling treatment at 100 F (37.8 C) shall be identical to that followed in the treatment at 77 F (25 C) (see 4.5.2.2), except that the temperature of both the reference standard acid and of the acid under test shall be brought to 100 F (38.7 C) prior to being added to the petri dishes containing the test panels.

The temperature shall then be maintained at 100 ± 2 F (1.1 C) for the 90 minute duration of the test. The panels shall be treated like those exposed at 77 F (25 C) and the average loss in weight of the two sets of panels calculated and compared. The panels shall then be observed under 6X magnification for evidence of pitting.

4.5.3 Hydrogen embrittlement-tensile effects.

4.5.3.1 Preparation of test specimens. Two 10-foot (3.05 meters) bars of 1213 cold drawn steel, 1/2 inch (1.3 cm) in diameter, shall be cut into 1-foot (0.3 meter) lengths. The steel bar used shall conform to QQ-S-637. The ten sections obtained from one bar shall be kept apart from the 10 sections of the other bar. The specimens shall be swabbed with naphtha, followed by acetone, washed with water, then with alcohol and air dried.

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4.5.3.2 Treatment procedure at 77 F (25 C). Eight of the specimens from each bar shall be placed into separate test tubes, 1-inch (2.5 cm) in diameter by 15 inches (38.1 cm) in length. The remaining two specimens of each bar shall be kept in reserve for control purposes. Four of the specimens of one of the bars shall be covered with the inhibited acid under test, and the other four specimens of that bar covered with the reference standard acid, with both acids at a temperature of 77 F (25 C). The specimens shall be allowed to remain in the acids, at a controlled temperature of 77 F ± 2 F (1.1 C) for 90 minutes. The specimens shall then be removed, washed thoroughly with water, dried with paper toweling, then washed with alcohol, and dried again with paper toweling. The average diameter of the individual specimens shall be measured half way between the ends. The specimens shall then be subjected to the tension test outlined in ASTM E 8, within one hour following the pickling treatment, using a gage length of 2 inches (5.1 cm). The control specimens reserved from the bar shall also be tested likewise. The smallest diameter at the point of break of each specimen shall be measured and the average reduction in area of the specimens pickled in the acid under test, of those pickled in the reference standard acid, and of the control specimens, calculated and compared for compliance with the requirements.

4.5.3.3 Treatment procedure at 100 F (37.8 C). The specimens obtained from the second bar shall be treated in similar manner as those treated at 77 F (25 C), except that the temperature of the acid under test and that of the reference standard acid shall be brought to 100 F (38.7 C) prior to being added to the test tube containing the specimens. The temperature shall be maintained at 100 F ± 2 F (1.1 C) for the 90-minute duration of the pickling treatment. The specimens shall be washed, dried, measured, broken, and their smallest diameter at the point of break determined in identical manner as for those specimens treated at 77 F (25 C). The control specimens shall be broken and measured in identical manner as the control specimens for the 77 F (25 C) test. The average reduction in area of the specimens pickled in the acid under test, of those pickled in the reference standard acid, and of the control specimens shall likewise be calculated and compared for compliance with the requrements.

4.5.4 Rate of rust removal.

4.5.4.1 Preparation of test panels. Twelve panels of cold-rolled and thoroughly annealed 18-gage 1020 steel, each measuring 2-1/2 by 1/2 inches (6.4 by 1.3 cm) shall be prepared. The steel sheets or strips used to make the panels shall conform to QQ-A-698, and shall have been exposed to the weather, under as uniform exposure conditions as possible, for a period of 6 months prior to the test.

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4.5.4.2 Procedure. The test panels shall be washed with petroleum ether and air-dried. Each panel shall be placed separately into test tubes of 25 millimeter diameter. A 12-ml portion of the acid under test, at 77 F (25 C), shall be added to each of six of the test tubes, and a like amount of the reference standard acid, also at 77 F (25 C), added to each of the other six test tubes, with the time of addition of the acids recorded. When the lower part of the panel is free of rust the panel shall be removed with tweezers, and inverted into the acid so that the rusted part of the panel is then at the bottom of the test tube. The time required for the rust to be removed from the two ends of the panels shall be recorded to the nearest minute, and the average rust-removing time for the two acids calculated and compared for compliance with the requirements.

4.6 Toxicological data and formulations. The contractor shall furnish the toxicological data and formulations required to evaluate the safety of the material for the proposed use.

5. PACKAGING

5.1 Packaging. The inhibited hydrochloric acid shall be packaged in containers of PPP-C-1337 in accordance with the requirements of level A, B, or Commerical. Government shipping regulations as stipulated in Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles (49 CPR 171-179) shall be complied with.

5.1.1 Level A - for air and surface shipments. The acid solution shall be packaged in metal containers with polyethylene liners conforming to PPP-C-1337, type II, class 1.

5.1.2 Level B - for surface shipments only. The acid solution shall be packaged in metal containers with polyethylene liners conforming to PPP-C-1337, type I, class 2.

5.1.3 Commercial. Packing shall be in accordance with MIL-STD-1188.

5.2 Packing. The 15-gallon (56.5 liter) metal containers with polyethylene liners shall be palletized in accordance with MIL-STD-147 except pallets may be unseasoned hardwood or softwood. Quantities less than a full pallet load will not be palletized.

5.3 Marking. In addition to the markings required by the Department of Transportation Regulations, shipments shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The inhibited hydrochloric acid is intended for removing heavy rust deposits from steel surfaces. It may also be used as a pickling acid employing rubber-lined steel tanks or wooden tanks for the pickling bath.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specificaton.
b. Size of container required if other than 15-gallon (see 5.1).
c. Level of packaging required (see 5.1).

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-H-13528B, "Hydrochloric Acid, Inhibited, Rust-Removing" FSC 6850

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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